PYTHON FOR PHYSICS / SPACED PRACTICE

A falling object's timestamps

10-15 minutes | NumPy + Matplotlib | No solution included

Start with 2 minutes writing the loop from memory. Then consult your previous code if needed. Focus on keeping time and velocity aligned.

PHYSICAL SETUP

An object is released from rest. Take downward as positive and ignore air resistance. Use constant gravitational acceleration:

dv/dt = g v(0) = 0 g = 9.81 m/s^2

YOUR TASK

1. Write simulate_velocity(dt, n_steps) using forward Euler. Return NumPy arrays of times and velocities, including the initial point. Keep the time variable local to the function.
2. Run 20 steps with dt = 0.1 s. Use an integer step counter to stop.
3. Print the first three (time, velocity) pairs and the final pair.
4. Plot velocity vertically against time horizontally, with units. Overlay the exact result v(t) = gt and print the maximum absolute error, comparing velocities at matching times.

PLAUSIBILITY CHECKS

Both arrays should contain 21 entries, start at (0, 0), and end at approximately t = 2 s. Each pair must describe the same instant.

ONE SENTENCE TO EXPLAIN

Unlike your RC simulation, should this Euler calculation have a noticeable discretization error? Explain using how acceleration changes with time.

NOTES

Submit your code and explanation for review before requesting a solution.